When uncertain, people use social prevalence information to update their beliefs. Does reliability of the information increase belief change?

I used Google and AI to find the research papers to support my hypotheses, check grammar, spelling errors and to interpret confidence intervals and other statistics.

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Abstract

Social media often presents information such as likes and comments, which may be interpreted as indicators of social prevalence, influencing belief formation and updating, especially under uncertainty. The present study examined whether initial belief certainty predicts belief change in response to social prevalence information and whether the perceived reliability of that information moderates belief updating. Participants (N = 27) rated their certainty about two ambiguous statements before and after viewing social prevalence information indicating that a majority of people believed the statements to be true. Belief change was calculated as the difference between final and initial belief certainty.

The results show a significant negative relationship between initial belief certainty and belief change. Participants with lower initial certainty updated their beliefs more, whereas participants with higher certainty showed less belief change. In contrast, perceived reliability of the social prevalence information did not significantly predict belief change and did not moderate the relationship between initial belief certainty and belief updating. Participants changed their beliefs to a similar extent regardless of whether the information was perceived as reliable or unreliable.

Introduction

There is a lot of misinformation spreading online these days. People are more likely to believe something to be true if the post has more likes or comments, mistaking social prevalence for truth. This happens because it is impossible to gather all the evidence or information related to a belief before forming it, which leads people to rely on those they trust when forming beliefs (Kruglanski, 2005). People also form beliefs based on others' opinions (Asch, 1951), sometimes without verifying the information, which further increases the spread of incorrect information. There are many factors influencing belief change, and not all beliefs are dependent on accuracy. Usually, an old belief is compared against a new belief, and change occurs when the new belief results in positive outcomes (Sharot, 2023).

We are uncertain about most of the new information we encounter on social media, which makes us more prone to believing fake news based on social prevalence. People are more likely to update their beliefs when it comes from a reliable source (Sanna, 2025). I hypothesized that even though people update their beliefs based on social prevalence when they are uncertain, the reliability of social prevalence information moderates belief change. I hypothesized that when people find social prevalence information reliable, they update their beliefs more, whereas when they find the information less reliable, they update their beliefs less.

When people are uncertain about something, they use social prevalence information to update their beliefs (Orticio, 2021). I expected to replicate past research and hypothesized that when people are uncertain about a belief or statement, they update their beliefs more. As the uncertainty reduces, people update their beliefs less.

Research has shown that the reliability of the source (Sanna, 2025) influences an individual's belief updating. In my study, I hypothesized that belief changes more when people perceive the information about a belief as reliable. Here, instead of asking participants whether the source of the information was reliable, they were asked about the reliability of the information itself.

Summary of Hypotheses

Hypothesis	**Null Hypothesis**	**Alternative**
Hypothesis 1 (Belief Change ~ Initial Belief Certainty)	People change beliefs equally when they are **both uncertain and certain**	People are more likely to change their beliefs **when they are uncertain**
Hypothesis 2 (Belief Change ~ Reliability of evidence)	The **reliability** of the **evidence does not influence** the belief change.	People change beliefs more when the **evidence is reliable**
Hypothesis 3 (Belief Change ~ Initial Belief Certainty + Reliability of Evidence)	People are more likely to update their beliefs based on **high social prevalence under uncertainty**	People are more likely to update their beliefs based on the <u>**reliability**</u> **of social prevalence information**

		under uncertainty

Methods

Participants

In this study, the survey was sent to students taking the Research Methods class by posting it on the class Discord channel and postbacc students' Slack channel. I also posted the study on Reddit's #takemysurvey channel and on Survey Circle. Participants were instructed to answer all the questions without using Google or AI. The study included a total of 20 questions including 2 attention check questions and 2 demographic questions, and could be completed within 4-5 minutes. There was no option available for skipping any question. The results from the class, Reddit's #takemysurvey channel, and Survey Circle were merged into one spreadsheet.

A total of 29 participants responded to the survey (12 men, 14 women, and 1 non-binary; mean age = 26.8). Two participants were excluded for failing at least one attention check, and the final data included 27 participants.

Procedures

In this study, participants first answered demographic questions about their age and gender. They were then shown the statement "Will Smith likes apples more than grapes" and were asked to answer three questions about how certain they are about the statement. Next, participants were shown a second statement, "the first baby born in the world in 1908 was a boy" and were asked to answer the same three certainty questions. Later, participants were shown social prevalence information ("7 out of 10 people thought the statement was true") about each statement and were asked to answer the same three questions about belief certainty. Participants were also asked two attention check questions, which required them to select a specific option. At the end, participants were asked whether they used Google or AI to answer any of the questions and whether they changed any of their answers. The final two questions about Google and AI were asked out of curiosity to see how much people used them to answer the survey questions.

Measures

Initial Belief Certainty

Participants were asked three questions about each of the two statements before and after viewing the social prevalence information. The three certainty questions for both statements were averaged to create an initial belief certainty score. Two of the questions measuring belief certainty were taken from a prior study (Orticio, 2021). These questions were "How likely do you think the statement is true?" and "How many people out of 10 do you think believe the statement is true?". The third question "How uncertain do you think the statement is true?" was designed to have three questions to measure belief certainty and was reverse-coded. Participants rated all questions on a scale from 0 to 10.

Belief Change

The three questions about certainty for both statements before viewing the social prevalence information were averaged to create an initial belief certainty score. After viewing the social prevalence information, participants answered the same three questions for each statement, and the scores of both statements were averaged to create a final belief certainty score. Belief Change was calculated as the difference between final belief certainty and initial belief certainty. In the original paper, the initial and final belief certainty were measured using the question "How likely do you think the statement is true?" separately for each statement, rather than using three questions and averaging the scores across statements as done in this study. The original study used multilevel modeling, which allowed belief certainty scores to be analyzed separately for each statement and therefore did not require combining scores across statements. In the present study, because a simple multiple regression model was used, the scores of all the statements were averaged together.

Reliability

Reliability was measured after participants viewed the social prevalence information. It was treated as a categorical variable with 2 levels - Reliable and Unreliable. The reliability ratings for both statements were merged such that if a participant rated the information as "Unreliable" for either the "Will Smith" or the "first boy" statement, the overall reliability classification was coded as "Unreliable". Participants were not assigned to any conditions. All participants answered the same survey questions and viewed the same social prevalence information.

Results

Descriptive Statistics

Descriptive statistics are reported below in Table 1.

Because belief change was calculated using initial and final belief certainty, the alpha reliability scores for both initial and final belief certainty were reported. The alpha reliability of initial belief certainty, which was calculated by averaging all three questions across both "Will Smith likes apples more than grapes" and "the first baby born in the world in 1908 was a boy" statements was 0.51. However, the alpha reliability for initial belief certainty calculated separately for the "Will Smith" and "first boy" statements was 0.74 and 0.71, respectively. Similarly, the alpha reliability of final belief certainty combining both "Will Smith" and "first boy" statements was 0.44. When calculated separately, the alpha reliability for final belief certainty was 0.52 for the "Will Smith" statement and 0.73 for the "first boy" statement.

The alpha reliability for each statement calculated separately was above 0.7 (except "Will Smith" final belief certainty) whereas the combined alpha reliability across both statements was 0.5 or below. This indicates that averaging the statements together to measure initial and final belief certainty was not an ideal approach and was done to accommodate the use of a multiple regression model. This represents one limitation of the study.

The mean belief change was 0.56, with a standard deviation of 0.66, meaning 68% of the participants' belief change was 0.66 above or below the mean belief change. Belief change ranged from -0.33 (indicating decreased belief in the statement) to 2 (indicating increased belief).

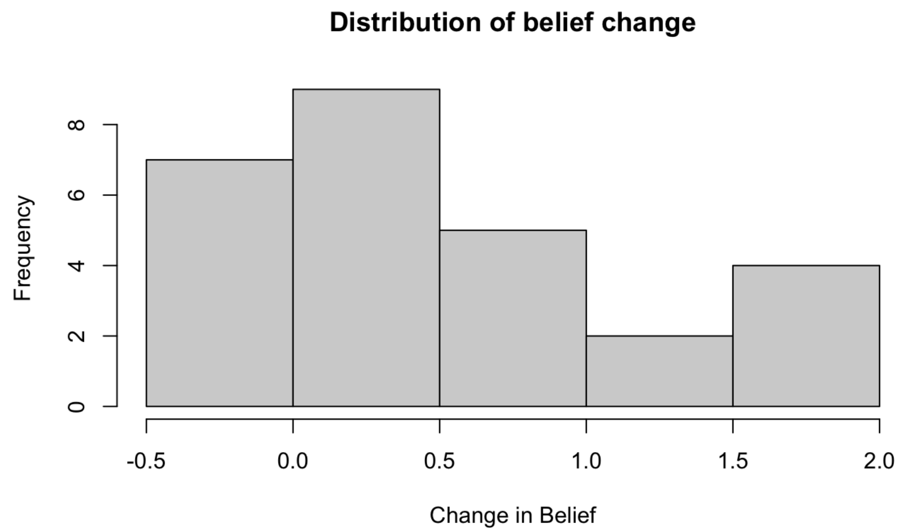
The mean initial belief certainty was 4.4, indicating that most participants' initial belief certainty was close to this value. Initial belief certainty ranged from 2.33 to 7.12, with a standard deviation of 1.12.

Ratings of the reliability of the social prevalence information were imbalanced, with 21 participants rating the information as "Unreliable" and only 6 participants rating it as "Reliable".

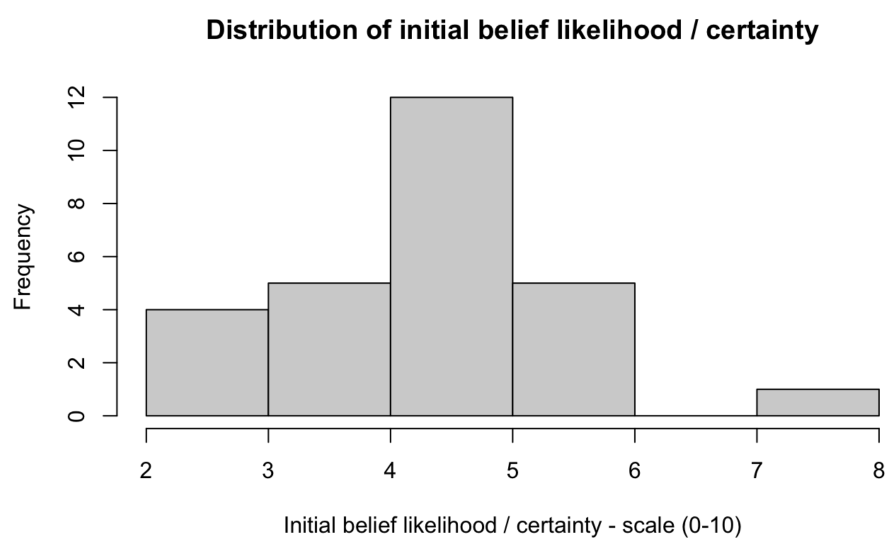
Table 1. Descriptive Statistics for Variables in Study

Variable	**Alpha Reliability**	**Mean**	**SD**	**Range**
Belief Change	Belief change = final belief - initial belief Alpha reliability of initial belief (combining Will Smith and Boy	0.5617284	0.6620806	-0.3333333 to 2.0000000

	scores) - 0.51 Alpha reliability of final belief (combining Will Smith and Boy scores) - 0.44			
Initial Certainty	Alpha reliability of initial belief (combining Will Smith and Boy scores) - 0.51	4.407407	1.123119	2.333333 to 7.166667
Reliability	Categorical variable	Reliable - 6 Unreliable - 21	-	-



Distribution of belief change
Frequency
0
2
4
6
8
-0.5
0.0
0.5
1.0
1.5
2.0
Change in Belief



Distribution of initial belief likelihood / certainty
Frequency
0
2
4
6
8
10
12
2
3
4
5
6
7
8
Initial belief likelihood / certainty - scale (0-10)

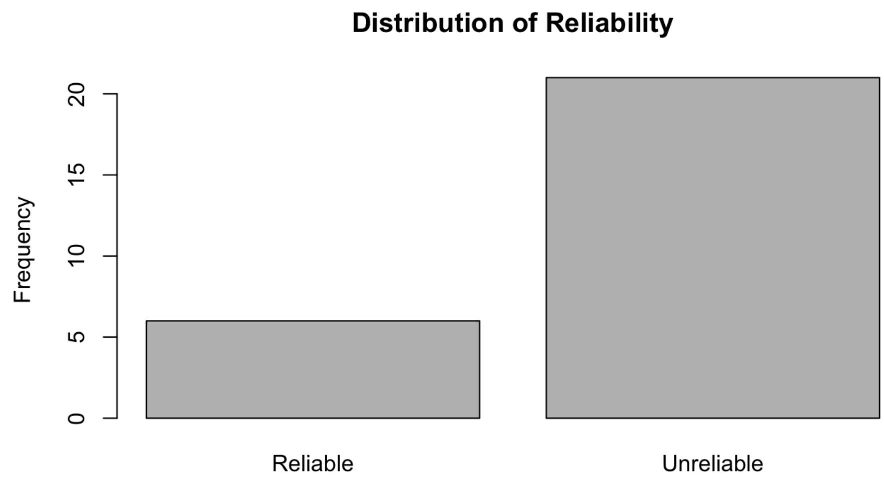


Figure 1. Distributions of Variables in Study

Linear Models and Inferential Statistics

Table 2. Predicting Shopping Habits from Mood and Moderation

DV = Belief Change	**Model 1** (Belief Change ~ Initial Certainty)	**Model 2** (Belief Change ~ Reliability)	**Model 3** (Belief Change ~ Initial Certainty + Reliability)
Intercept	*1.5974*	*0.55*	*1.7526*
Initial Belief Certainty	*-0.235*	--	*-0.2463*
Reliability	--	*0.0079*	*-0.1357*

R^2	**0.1589**	**0.00002**	**0.1661**

*Note : * p < .05, ** p < .01. Number in parentheses are standard errors, estimated using Null Hypothesis Significance Testing.*

Hypothesis 1. Predicting Belief Change from Initial Belief Certainty

I hypothesised that when people are uncertain about a statement, they are more likely to change their belief. The initial belief certainty predicts belief change, such that as certainty about a statement increases, belief change reduces. The slope of -.0235 indicates a negative relationship between initial belief certainty and belief change. As initial belief certainty increases by one unit, the change in belief reduces by 0.235 units, with a 95% confidence interval of [-0.457, -0.012]. The t-value of -2.173 shows that the slope is two times smaller than the standard error if the null hypothesis was true. The relationship is statistically significant with a p-value of 0.03, meaning that the probability of finding this relationship if the null hypothesis was true is less than 5%. R-squared value of 0.158 indicates that 15.8% of variation in belief change is explained by initial belief certainty. Overall, the data supports the hypothesis that change in belief decreases as certainty about a statement increases.

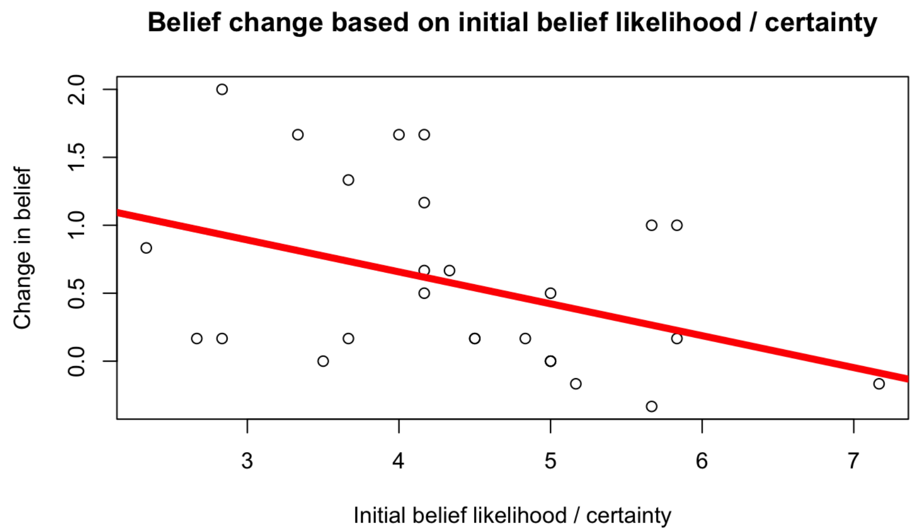


Figure 2. Graph of Model 1

Hypothesis 2. Predicting Belief Change from Reliability of social prevalence information

I hypothesized that when people find the social prevalence information reliable, they update their beliefs more whereas when they find the information less reliable, they update their beliefs less. Thus, the reliability of the social prevalence information was expected to predict belief change.

The mean belief change was 0.55 when participants perceived the social prevalence information as unreliable. When participants thought that the information was reliable, the change in belief was around 0.56. The difference between the reliable and unreliable conditions was very small, indicating that participants who thought the information was reliable and those who thought the information was unreliable changed their beliefs similarly. The 95% confidence interval was [-0.635, 0.651]. The p-value of 0.979 indicates a high probability of finding this relationship if the null hypothesis were true, and because the 95% confidence interval includes zero, the results were not statistically significant. The R-squared value is too low to explain any variation in belief change. Overall, the data did not support the hypothesis that participants would change their beliefs more when the information was perceived as reliable.

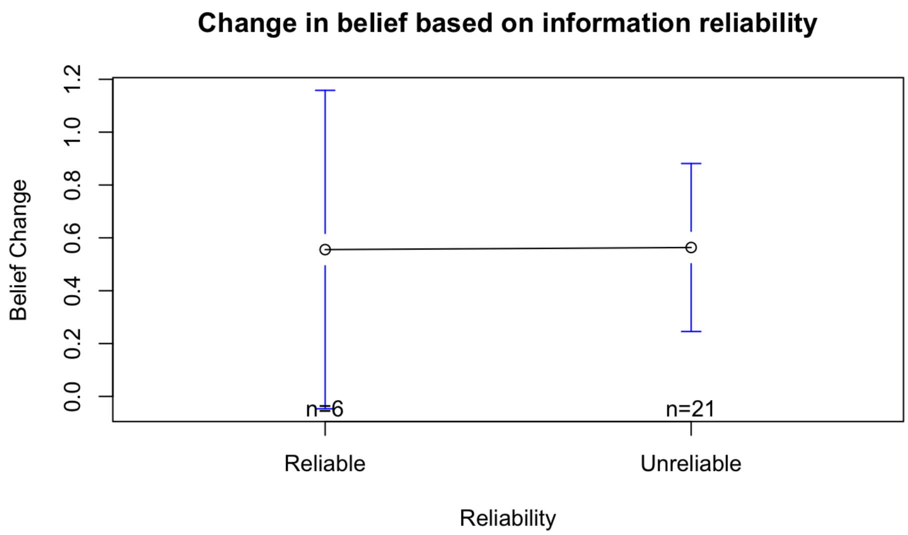


Figure 3. Graph of Model 2

Hypothesis 3. Predicting Belief Change from Initial Belief Certainty and Reliability of Social Prevalence Information

I hypothesised that belief change would be greater when individuals were more uncertain about a statement. Initial belief certainty was expected to predict belief change, such that higher certainty would be associated with less belief change. The new slope of -.0246 with reliability added to the model, shows that there is a negative relationship between initial belief certainty and belief change as before. As initial belief certainty increases by 1 unit, the change in belief reduces by 0.246 units, with the 95% confidence interval of [-0.478, -0.013]. The t-value of -2.186 shows that the slope is two times smaller than the standard error if the null hypothesis was true. The relationship is statistically significant with a p-value of 0.03, meaning that the probability of finding this relationship if the null hypothesis was true is less than 5%. Overall, the data supports the hypothesis that belief change decreases as certainty about a statement increases.

Adding reliability of the social prevalence information to the model increased the negative relationship between initial belief certainty and belief change by a very small amount (0.01). Even though the belief change in reliable condition was 0.135 less than the unreliable condition, the relationship between reliability and belief change remained insignificant, with a p-value of 0.65 and a 95% confidence interval of [-0.752, 0.48]. The combined model has an R-squared value of 0.166 which means that 16.6% of the variation in belief change is explained by initial belief certainty and reliability of social prevalence information together.

Overall, the reliability of the social prevalence information did not moderate the relationship between initial belief certainty and belief change, with belief change remaining similar when participants perceived the information as reliable or unreliable.

Discussion

Summary of the Current Study

The study examined how initial belief certainty and the perceived reliability of social prevalence information influence belief change. Participants reported their certainty about two statements before and after viewing social prevalence information indicating that a majority of people believed the statements to be true. The results showed a significant effect of initial belief certainty on belief change. When initial belief certainty was low, participants showed greater belief updating, and as belief certainty increased, the magnitude of belief change decreased. These findings suggest that uncertainty plays a key role in how individuals use social prevalence information to update their beliefs.

In contrast, the perceived reliability of the social prevalence information did not significantly predict belief change, nor did it moderate the relationship between initial belief certainty and belief change. Although it was hypothesized that belief updating would be greater when social prevalence information was perceived as reliable, participants updated their beliefs similarly regardless of whether the information was judged to be reliable or unreliable. Overall, the theories tested in this study were partially supported. The results align with prior research demonstrating that initial belief certainty predicts belief change under high social prevalence conditions, but the role of perceived information reliability in belief updating could not be established in this study.

Limitations and Future Directions

There are several limitations to this study. First, the sample size was small, which limits the generalizability of the findings. Second, the study used only two statements, “Will Smith likes apples more than grapes” and “the first baby born in the world in 1908 was a boy”, both of which are relatively ambiguous or uncertain in nature. The absence of more certain control statements (e.g., “Plants need water to grow”) limits the ability to compare belief updating across different levels of baseline certainty. Finally, certainty and reliability scores across all statements were aggregated into a single measure. Future research should address these limitations by using larger samples, including both certain and uncertain statements, and analyzing belief change at the statement level.

References

Asch, S. E. (1951). Effects of group pressure upon the modification and distortion of judgments.

Kruglanski, A. W., Raviv, A., Bar-Tal, D., Raviv, A., Sharvit, K., Ellis, S., & Mannetti, L. (2005). Says who? Epistemic authority effects in social judgment. Advances in experimental social psychology, 37(37), 345-92.

Orticio, E., Marti, L., & Kidd, C. (2021). Beliefs are most swayed by social prevalence under uncertainty. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 43, No. 43).

Sanna, G. A., & Lagnado, D. (2025). Belief updating in the face of misinformation: The role of source reliability. *Cognition*, *258*, 106090.

Sharot, T., Rollwage, M., Sunstein, C. R., & Fleming, S. M. (2023). Why and when beliefs change. Perspectives on Psychological Science, 18(1), 142-151.

Supplemental Materials

Link to Code and Data

Link to Survey

Code

```
---

title: "Final Project Milestone 5"

format: html

editor: visual

author: "Swetha Srikari"

---
```

Loading Data

```
```{r load}

data <- read.csv('../datasets/project/project_data.csv', stringsAsFactors = T)
```
```

Cleaning Data

````
```{r remove timestamp column}


data <- data[, -c(1,1)]
head(data)
```


```{r creating a codebook}


code_names <- c('age', 'gender',
        "How likely do you think Will Smith likes grapes more than apple is true?",
        "How many people out of 10 do you think believe that Will Smith likes grapes more than apples is true?",
        "How uncertain are you that Will Smith likes grapes more than apples is true?",
        "Please choose the option Haha for this question",
        "How likely do you think the first baby born in the world in 1908 was a boy is true?",
        "How many people out of 10 do you think believe that the first baby born in the world in 1908 was a boy is true?",
        "How uncertain are you that the first baby born in the world in 1908 was a boy is true?",
        "How reliable do you think the information about Will Smith is?",
        "How likely do you think it is that Will Smith likes grapes more than apples is true?",
        "How uncertain are you that Will Smith likes grapes more than apples is true?",
````
````

"How many people out of 10 do you think believe that Will Smith likes grapes more than apples is true?",

"Please choose 67 for this question.",

"How reliable do you think the information about the first boy born in 1908 is?",

"How likely do you think that the first baby born in the world in 1908 was a boy is true?",

"How uncertain are you that the first baby born in the world in 1908 was a boy is true?",

"How many people out of 10 do you think believe that the first baby born in the world in 1908 was a boy is true?",

"Did you go back to change any of your answers?",

"Did you use Google or AI to answer any of the questions?",

"Where did you collect the data?")

codebook <- data.frame(names(data), code_names)

write.csv(codebook, "Desktop/101/datasets/project/project_codebook.csv")

```

```{r remove data points with at least 1 attention check fail}

data <- data[!(data$attention.check1 == "Hehe" | data$attention.check2 == 57), ]

#check for missing attention checks

data[data$attention.check1 == "Hehe" | data$attention.check2 == 57, ] # no data points with attention check fail

```

- the first 6-rows of your data as loaded into R (e.g., head(d))

    ```{r head}
    head(data)
    ```

- a histogram of one of your numeric variables.

    ```{r histogram}

    hist(data$will.likely.before,
         main="How likely do you think Will Smith likes grapes more than apple is true?",
         xlab="Scale:1-10")
    ```

```{r age}

hist(data$age)
mean(data$age)

```

```{r gender}

plot(data$gender)
table(data$gender)
```

Milestone 5

Part 1

Independent variable - Initial Belief likelihood/certainty (how much people believe the statement is true): Belief likelihood, belief prevalence, belief uncertainty (reverse coded) before watching prevalence information

Checking consistency for Will Smith question alone - 0.74

```{r checking consistency of IV for Will Smith question}
will_before <- data.frame(data$will.likely.before, data$will.prevalence.before,
10-data$will.uncertain.before)

library(psych)

psych::alpha(will_before)

```

Checking consistency for boy question alone 0.71

```{r boy before consistency}

boy_before <- data.frame(data$boy.likely.before, data$boy.prevalence.before,

10-data$boy.uncertain.before)

library(psych)

psych::alpha(boy_before)

```

Final belief likelihood/ certainty: Belief likelihood, belief prevalence, belief uncertainty (reverse coded) after watching prevalence information

Checking consistency for Will question alone 0.52

```{r will after consistency}

will_after <- data.frame(data$will.likely.after, data$will.prevalence.after,

10-data$will.uncertain.after)

library(psych)

```
psych::alpha(will_after)
```

Checking consistency for boy question alone - 0.73

```{r boy after consistency}
boy_after <- data.frame(data$boy.likely.after, data$boy.prevalence.after,
10-data$boy.uncertain.after)

library(psych)
psych::alpha(boy_after)
```

Combining Will Smith and boy scores for IV

Initial Belief likelihood / certainty (how much people believe the statement is true): Belief likelihood, belief prevalence, belief uncertainty (reverse coded) before watching prevalence information - 0.51

```{r}
belief_before.df <- data.frame(data$will.likely.before, data$boy.likely.before,
                              data$will.prevalence.before, data$boy.prevalence.before,
                              10-data$will.uncertain.before, 10-data$boy.uncertain.before)
```

library(psych)

psych::alpha(belief_before.df)

```

**Final belief likelihood / certainty** (how much people believe the statement is true after watching the info)**:** Belief likelihood, belief prevalence, belief uncertainty (reverse coded) after watching prevalence information 0.44

```{r}
belief_after.df <- data.frame(data$will.likely.after, data$boy.likely.after,
                              data$will.prevalence.after, data$boy.prevalence.after,
                              10-data$will.uncertain.after, 10-data$boy.uncertain.after)

library(psych)
psych::alpha(belief_after.df)
```

Initial Belief and Final Belief

```{r}
data$initial_belief <- rowMeans(belief_before.df, na.rm = T)

````
data$final_belief <- rowMeans(belief_after.df, na.rm = T)
```

Change in belief


```{r}
data$belief_change <- data$final_belief - data$initial_belief
```


Descriptive statistics

DV - belief change


```{r DV - belief change}
hist(data$belief_change,
    xlab="Change in Belief",
    main = "Distribution of belief change")
mean(data$belief_change)
sd(data$belief_change)
range(data$belief_change)
summary(data$belief_change)
```
````
````

IV1 - initial belief likelihood / certainty

~~~
```{r IV1 - initial belief}
hist(data$initial_belief,
    xlab = "Initial belief likelihood / certainty - scale (0-10)",
    main = "Distribution of initial belief likelihood / certainty")
mean(data$initial_belief)
sd(data$initial_belief)
range(data$initial_belief)
summary(data$initial_belief)
```
~~~

If the answer to 'how reliable do you think the info is?' is 'unreliable' for either Will Smith or boy question, it is considered 'unreliable'

~~~
```{r}
#check how many different
sum(data$will.reliable == data$boy.reliable)
nrow(data)

data[data$will.reliable == "Unreliable" | data$boy.reliable == "Unreliable", 'reliability'] =
"Unreliable"
~~~

data[!(data$will.reliable == "Unreliable" | data$boy.reliable == "Unreliable"), 'reliability'] = "Reliable"

sum(data$reliability == "Reliable")
```

IV2 - reliability

```{r IV2 - reliability}
data$reliability <- as.factor(data$reliability)
table(data$reliability)
plot(data$reliability,
    ylab="Frequency",
    main="Distribution of Reliability")
```

Part 2: Inferential and Predictive Statistics

Model 1: belief change \~ initial belief + error

```{r model 1}

mod1 <- lm(data$belief_change ~ data$initial_belief)

~~~~
plot(data$belief_change ~ data$initial_belief,
     xlab = "Initial belief likelihood / certainty",
     ylab = "Change in belief",
     main = "Belief change based on initial belief likelihood / certainty")
abline(mod1, lwd=5, col='red')

summary(mod1)
confint(mod1, level = 0.95)
```

Model 2: belief change \~ reliability + error

```{r model 2}

mod2 <- lm(data$belief_change ~ data$reliability)
plot(data$belief_change ~ data$reliability,
     xlab = "Reliability",
     ylab = "Belief Change",
     main = "Change in belief based on information reliability")

library(gplots)

gplots::plotmeans(data$belief_change ~ data$reliability,
~~~~

xlab = "Reliability",

ylab = "Belief Change",

main = "Change in belief based on information reliability")

summary(mod2)

confint(mod2)

```

Model 3: belief change \~ initial belief likelihood + reliability + error

```{r model 3}

mod3 <- lm(data$belief_change ~ data$initial_belief + data$reliability)

summary(mod3)

confint(mod3)

```
```